Example 1

```
supplement_heart_rate |> anovas(outcome = resting_heart_rate_bpm,
                                A = treatment,
                                B = pony_type,
                                interaction = TRUE)
```

```
Two-way ANOVA (with interaction)

Source               df         SS        MS       F  p-value
──────────────────────────────────────────────────────────────
treatment             1   215.2813  215.2813  15.409  < 0.001
pony_type             2  1383.6803  691.8401  49.519  < 0.001
treatment:pony_type   2   636.6508  318.3254  22.785  < 0.001
Residuals            66   922.0942   13.9711
──────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(2, 66) = 22.785
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 2

```
exercise_recovery |> anovas(outcome = recovery_time_min,
                            A = treatment,
                            B = pony_type,
                            interaction = TRUE)
```

```
Two-way ANOVA (with interaction)

Source               df         SS         MS       F  p-value
──────────────────────────────────────────────────────────────────
treatment             2  2909.4913  1454.7456  79.154  < 0.001
pony_type             2    23.1424    11.5712   0.630    0.535
treatment:pony_type   4    47.5704    11.8926   0.647    0.630
Residuals            99  1819.4858    18.3786
──────────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(4, 99) = 0.647
p-value: 0.630
Conclusion: Fail to reject the null hypothesis (p = 0.630 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

```
exercise_recovery |> anovas(outcome = recovery_time_min,
                            A = treatment,
                            B = pony_type,
                            interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source      df         SS         MS       F  p-value
──────────────────────────────────────────────────────
treatment    2  2909.4913  1454.7456  80.254  < 0.001
pony_type    2    23.1424    11.5712   0.638    0.530
Residuals  103  1867.0562    18.1268
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
Test statistic: F(2, 103) = 80.254
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 103) = 0.638
p-value: 0.530
Conclusion: Fail to reject the null hypothesis (p = 0.530 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice 1

Applejack and the Ponyville clinic are studying whether a high-fiber feed changes postmeal blood glucose. Within each pony type, adult ponies are randomly assigned to Standard feed or High-fiber feed for two weeks. After a standardized meal, glucose is measured once at the same pre-specified time for every pony.

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,
                  A = treatment,
                  B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source               df         SS         MS       F  p-value
──────────────────────────────────────────────────────────────────
treatment             1   531.4410   531.4410  13.231  < 0.001
pony_type             2  2111.6527  1055.8263  26.286  < 0.001
treatment:pony_type   2  2635.6647  1317.8323  32.809  < 0.001
Residuals            84  3374.0227    40.1669
──────────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(2, 84) = 32.809
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,
                  A = treatment,
                  B = pony_type,
                  interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source     df         SS         MS       F  p-value
──────────────────────────────────────────────────────
treatment   1   531.4410   531.4410   7.605    0.007
pony_type   2  2111.6527  1055.8263  15.109  < 0.001
Residuals  86  6009.6873    69.8801
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
```

```
H₁: At least one mean across levels of treatment differs
Test statistic: F(1, 86) = 7.605
p-value: 0.007
Conclusion: Reject the null hypothesis (p = 0.007 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 86) = 15.109
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 2

Fluttershy is testing a fictional anti-inflammatory treatment in adult ponies with the same mild inflammatory condition. Within each pony type, ponies are randomly assigned to Placebo, Low dose, or High dose. A blood inflammation marker is measured once after seven days.

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,
                       A = treatment,
                       B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                df        SS        MS       F  p-value
──────────────────────────────────────────────────────────────
treatment              2  384.4924  192.2462  86.419  < 0.001
pony_type              2  366.1000  183.0500  82.286  < 0.001
treatment:pony_type    4  303.1356   75.7839  34.067  < 0.001
Residuals            126  280.2960    2.2246
──────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(4, 126) = 34.067
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,
                       A = treatment,
                       B = pony_type,
                       interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source      df        SS        MS       F  p-value
──────────────────────────────────────────────────────
treatment    2  384.4924  192.2462  42.836  < 0.001
pony_type    2  366.1000  183.0500  40.787  < 0.001
Residuals  130  583.4316    4.4879
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
```

```
Test statistic: F(2, 130) = 42.836
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 130) = 40.787
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 3

Nurse Redheart compares dressings in adult ponies receiving care for comparable minor wounds. Within each pony type, ponies are randomly assigned to Standard dressing, Hydrogel, or Enhanced hydrogel. The clinic records the time from first dressing to complete wound closure using a common clinical criterion. Each pony contributes one wound.

```
wound_healing |> anovas(outcome = wound_closure_days,
                        A = treatment,
                        B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                df         SS        MS        F  p-value
──────────────────────────────────────────────────────────────────
treatment              2  1306.4801  653.2401  241.137  < 0.001
pony_type              2   944.2704  472.1352  174.284  < 0.001
treatment:pony_type    4     6.6327    1.6582    0.612    0.655
Residuals            126   341.3333    2.7090
──────────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(4, 126) = 0.612
p-value: 0.655
Conclusion: Fail to reject the null hypothesis (p = 0.655 ≥ α = 0.050)
```

```
wound_healing |> anovas(outcome = wound_closure_days,
                        A = treatment,
                        B = pony_type,
                        interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source      df         SS        MS        F  p-value
──────────────────────────────────────────────────────
treatment    2  1306.4801  653.2401  244.050  < 0.001
pony_type    2   944.2704  472.1352  176.390  < 0.001
Residuals  130   347.9661    2.6767
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
```

```
H₁: At least one mean across levels of treatment differs
Test statistic: F(2, 130) = 244.05
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 130) = 176.39
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).

Let's Practice 4

Rainbow Dash helps the Ponyville clinic compare two fluid treatments after a standardized exercise session. Within each pony type, adult ponies are randomly assigned to Standard fluids or Electrolyte fluids. Time to reach a prespecified hydration target is recorded once for each pony.

```
rehydration |> anovas(outcome = rehydration_time_min,
                      A = treatment,
                      B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source               df         SS         MS        F  p-value
──────────────────────────────────────────────────────────────────
treatment             1  2301.2890  2301.2890  258.932  < 0.001
pony_type             2    36.5502    18.2751    2.056    0.134
treatment:pony_type   2     2.4427     1.2213    0.137    0.872
Residuals            84   746.5613     8.8876
──────────────────────────────────────────────────────────────────


Hypotheses (treatment:pony_type):
H₀: There is no interaction between treatment and pony_type
H₁: There is an interaction between treatment and pony_type
Test statistic: F(2, 84) = 0.137
p-value: 0.872
Conclusion: Fail to reject the null hypothesis (p = 0.872 ≥ α = 0.050)
```

```
rehydration |> anovas(outcome = rehydration_time_min,
                      A = treatment,
                      B = pony_type,
                      interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source     df         SS         MS        F  p-value
──────────────────────────────────────────────────────
treatment   1  2301.2890  2301.2890  264.232  < 0.001
pony_type   2    36.5502    18.2751    2.098    0.129
Residuals  86   749.0040     8.7093
──────────────────────────────────────────────────────


Hypotheses (treatment):
H₀: All means across levels of treatment are equal
H₁: At least one mean across levels of treatment differs
```

```
Test statistic: F(1, 86) = 264.232
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 86) = 2.098
p-value: 0.129
Conclusion: Fail to reject the null hypothesis (p = 0.129 ≥ α = 0.050)
```

1. Are you reporting the interaction or main effects? Why?

2. Report the appropriate hypothesis test(s).